

EXERCISE 1

INTRODUCTION TO CRYPTOGRAPHY AND DATA SECURITY

1. We consider a cryptosystem where users enter keys in the form of passwords. Assume a password consisting of 8 letters, where each letter is encoded with the ASCII code (7 bits per character, i.e., 128 possible characters).

- (1) What is the size of the key space that can be constructed with such passwords?
- (2) What is the corresponding key length in bits?
- (3) Assuming most users only use the 26 lowercase letters from the alphabet instead of the full 7 bits of the ASCII encoding, what is the corresponding key length in bits in this case?
- (4) What is the minimum number of characters needed for a password to generate a key length of 128 bits, assuming:
 - (a) 7-bit characters?
 - (b) 26 lowercase letters from the alphabet?

2. Let X be a uniform random variable over the set $\{0, 1\}^n$. Let Y be an arbitrary random variable over the set $\{0, 1\}^n$ (not necessarily uniform) that is independent of X . Define the random variable $Z = X \oplus Y$. What is the probability that Z equals 0^n ?

3. Assume you know the ciphertext of the message “attack at dawn” using one-time pad encryption is

6c73d5240a948c86981bc294814d

(the plaintext is in 8-bit ASCII format and the ciphertext is written in hexadecimal). What is the ciphertext of the message “attack at dusk” with the same OTP key?

4. Modular arithmetic is fundamental to many cryptosystems. We will practice some modular computations:

Compute the following results without a calculator:

- (1) $15 \cdot 29 \bmod 13$
- (2) $2 \cdot 29 \bmod 13$
- (3) $2 \cdot 3 \bmod 13$
- (4) $-11 \cdot 3 \bmod 13$

5. Compute the following without a calculator:

- (1) $1/5 \bmod 13$
- (2) $1/5 \bmod 7$
- (3) $2/5 \bmod 7$

6.

- (1) Construct the addition and multiplication tables for $\mathbb{Z}_4$.
- (2) Construct the addition and multiplication tables for $\mathbb{Z}_5$.
- (3) Construct the addition and multiplication tables for $\mathbb{Z}_6$.
- (4) Identify the elements in $\mathbb{Z}_4$ and $\mathbb{Z}_6$ that lack a multiplicative inverse. Explain why a multiplicative inverse exists for all nonzero elements in $\mathbb{Z}_5$.

7. What are the multiplicative inverses of 5 in $\mathbb{Z}_{11}$, $\mathbb{Z}_{12}$, and $\mathbb{Z}_{13}$? You can use trial-and-error with a calculator or computer.

This simple problem highlights the crucial fact that the inverse of an integer in a given ring depends entirely on the modulus. The extended Euclidean algorithm, introduced in Section 6.3, helps compute inverses efficiently.

8. Compute x as far as possible without a calculator. Use a smart decomposition of the exponent as shown in the example in Section 1.4.1:

- (1) $x = 3^2 \pmod{13}$
- (2) $x = 7^2 \pmod{13}$
- (3) $x = 3^{10} \pmod{13}$
- (4) $x = 7^{100} \pmod{13}$
- (5) $7^x = 11 \pmod{13}$

The last problem is a discrete logarithm, a hard problem discussed in Chapter 8. The security of many public-key schemes relies on the difficulty of solving the discrete logarithm for large numbers, typically over 2000 bits.

9. Find all integers n with $0 < n < m$ that are relatively prime to m , for $m = 4, 5, 9, 26$. The number of these integers is denoted by $\phi(m)$, e.g., $\phi(3) = 2$. This function is called "Euler" 's phi function." What is $\phi(m)$ for $m = 4, 5, 9, 26$?

Further information on Euler's phi function will be discussed in Section 6.3.

10. This problem deals with the affine cipher where the key is given as $a = 7$ and $b = 22$.

(1) Decrypt the text below:

falszztysyzyjkywjrztyjztyynaryjkswarztyegyyj

(2) Who wrote the line?

11. An obvious approach to increase the security of a symmetric algorithm is to apply the same cipher twice, i.e.,

$$y = e_{k_2}(e_{k_1}(x))$$

As is often the case in cryptography, things can be tricky and results are often different from the expected or desired ones. In this problem, we show that a double encryption with the affine cipher is only as secure as single encryption! Assume two affine ciphers $e_{k_1} \equiv a_1x + b_1 \pmod{26}$ and $e_{k_2} \equiv a_2x + b_2 \pmod{26}$.

(1) Show that there is a single affine cipher $e_{k_3} \equiv a_3x + b_3 \pmod{26}$ which performs exactly the same encryption (and decryption) as the combination $e_{k_2}(e_{k_1}(x))$.

(2) Find the values for a_3, b_3 when $a_1 = 3, b_1 = 5$ and $a_2 = 11, b_2 = 7$.

(3) To verify your solution,

(a) Encrypt the letter K with e_{k_1} and the result with e_{k_2} .

(b) Encrypt the letter K with e_{k_3} .

(4) Briefly describe what happens if an exhaustive key-search attack is applied to a double-encrypted affine ciphertext. Is the effective key space increased?

Remark: The issue of multiple encryption is of great practical importance in the case of the Data Encryption Standard (DES), for which multiple encryption (in particular, triple encryption) does increase security considerably, cf. Section 5.3.2.

12. We already know that the substitution cipher and the shift cipher can easily be broken in little time. Let us now consider an extension of the shift cipher, namely the Vigenère cipher (named after Blaise de Vigenère). Instead of using a single key k for the shift, it uses l different shifts that are derived from a secret code word c . The code word consists of l letters and has the form $c = (c_0, c_1, \dots, c_{l-1})$. Each letter c_i corresponds to a number $0, \dots, 25$, which is given by its position in the alphabet. These numbers are the l shift positions, which we denote by $(k_0, k_1, \dots, k_{l-1})$.

Encryption (and decryption) work as follows: The first plaintext letter x_0 is cyclically shifted by k_0 positions, the second plaintext letter x_1 by k_1 positions, and so on, until plaintext letter x_{l-1} is shifted by k_{l-1} positions. From now on, the shift sequence repeats, i.e., plaintext x_l is again shifted by k_0 positions, the next plaintext by k_1 positions, and so on. This process is expressed as:

$$y_j \equiv x_j + k_{(j \bmod l)} \pmod{26}$$

where x_j denotes the j -th letter of the plaintext $x = (x_0, x_1, \dots)$.

Since the cipher uses many ciphertext alphabets, it is called a polyalphabetic cipher.

(1) Assume the code word is given as $c = \text{JAMAICA}$ of size $l = 7$. Transform the code word into the corresponding encryption keys k_i . You can use Table 1 for this task.

- (2) Use the table to encrypt the word $x = \text{CODEBREAKERS}$ with the Vigenère cipher. For each plaintext letter, choose the row with the corresponding shift value in the leftmost column # and look up the shifted version of the plaintext.
- (3) What do you think about the security of the Vigenère cipher? Propose an attack.

#	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
0	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
1	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
2	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
3	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
4	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
5	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
6	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
7	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
8	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
9	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
10	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
11	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
12	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
13	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
14	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
15	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
16	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
17	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
18	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
19	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
20	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
21	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
22	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
23	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
24	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
25	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

TABLE 1. Polyalphabetic substitution table